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(54) **HYBRID SCHEDULING FOR HETEROGENEOUS PROCESSOR SYSTEMS**

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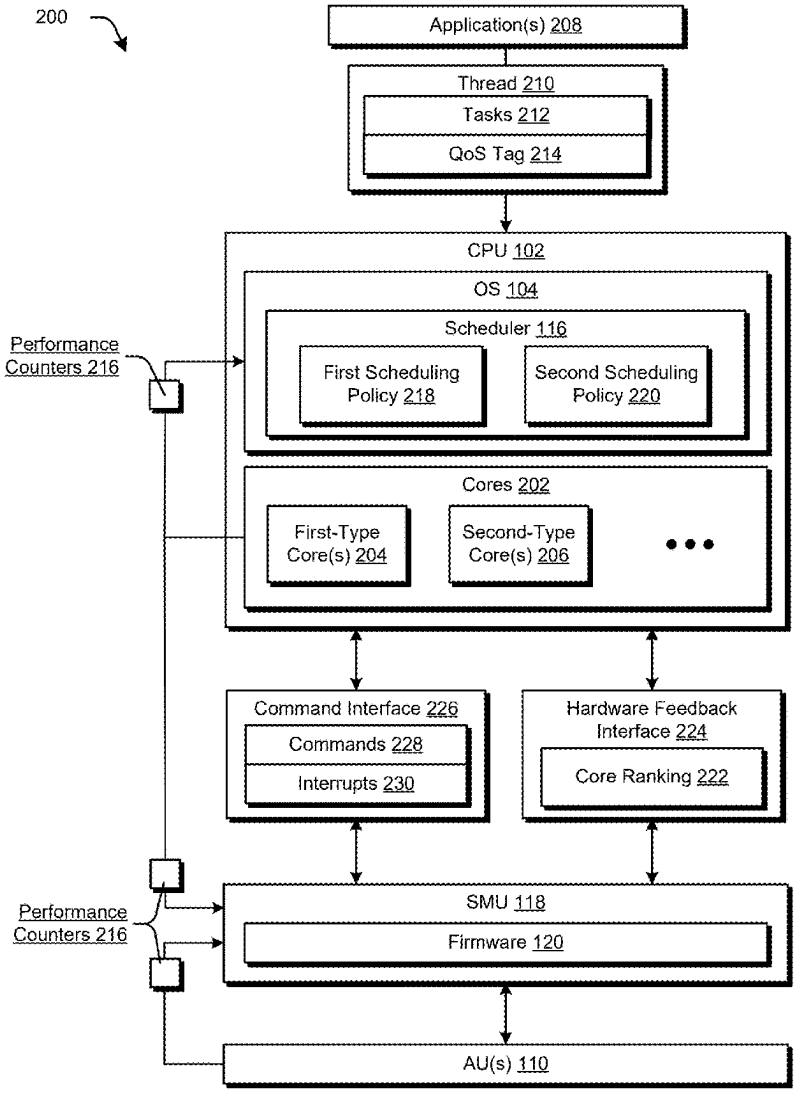
(57) **ABSTRACT**

Hybrid scheduling for heterogeneous processor systems is described. In one or more implementations, a system includes a central processing unit having multiple cores of at least two different core types, one or more accelerator processors, and a system management processor. The system management processor is configured to update a scheduling policy implemented by an operating system of the central processing unit from a first scheduling policy to a second scheduling policy based on a utilization of the multiple cores and/or the one or more accelerator processors. The update enables the system management processor to control task scheduling.



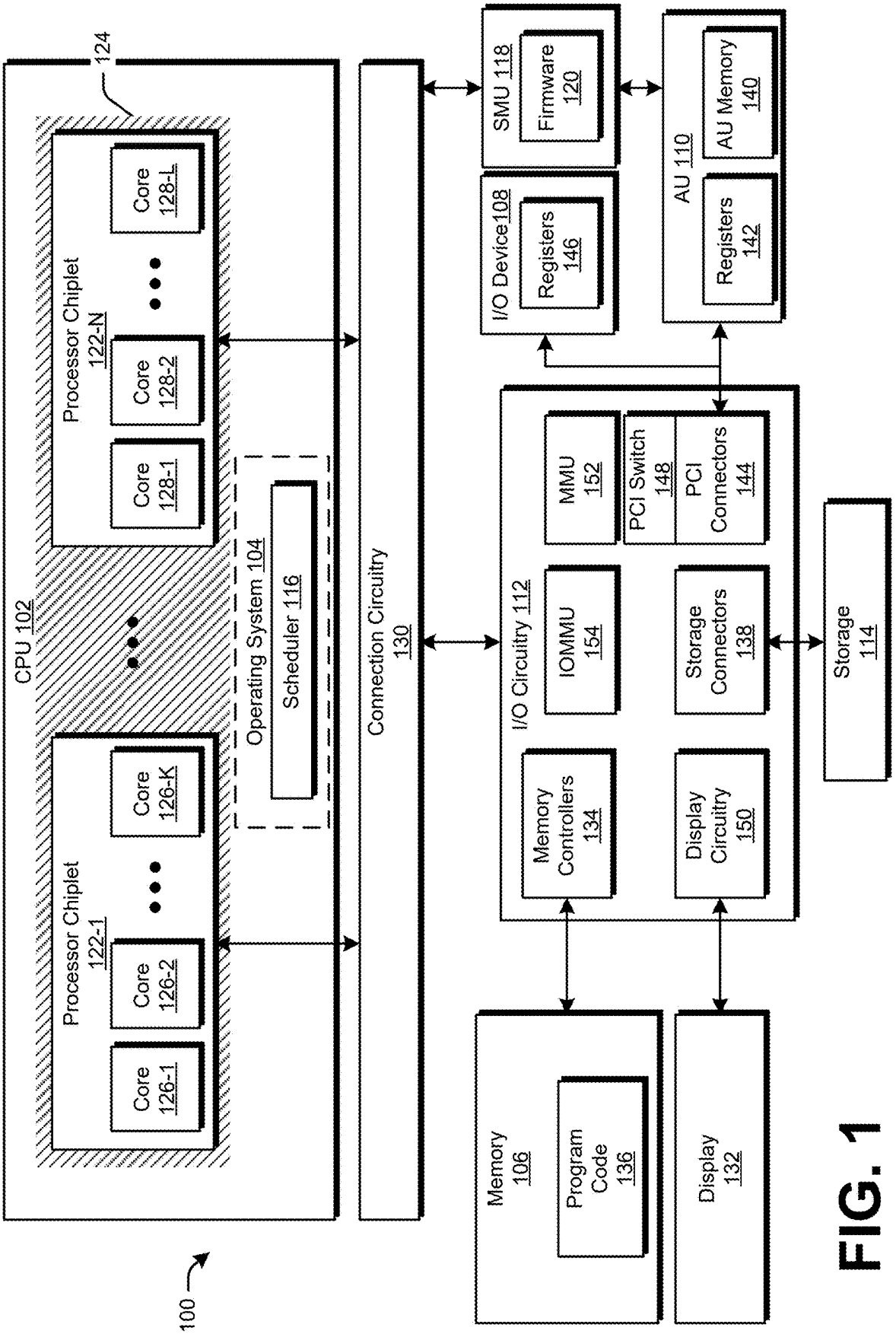

100
CPU 102
124
Processor Chiplet 122-N
Core 128-L
Core 128-2
Core 128-1
Processor Chiplet 122-1
Core 126-K
Core 126-2
Core 126-1
Operating System 104
Scheduler 116
Connection Circuitry 130
SMU 118
Firmware 120
I/O Device 108
Registers 146
AU 110
AU Memory 140
Registers 142
I/O Circuitry 112
MMU 152
PCI Switch 148
PCI Connectors 144
IOMMU 154
Storage Connectors 138
Storage 114
Memory Controllers 134
Display Circuitry 150
Memory 106
Program Code 136
Display 132

FIG. 1

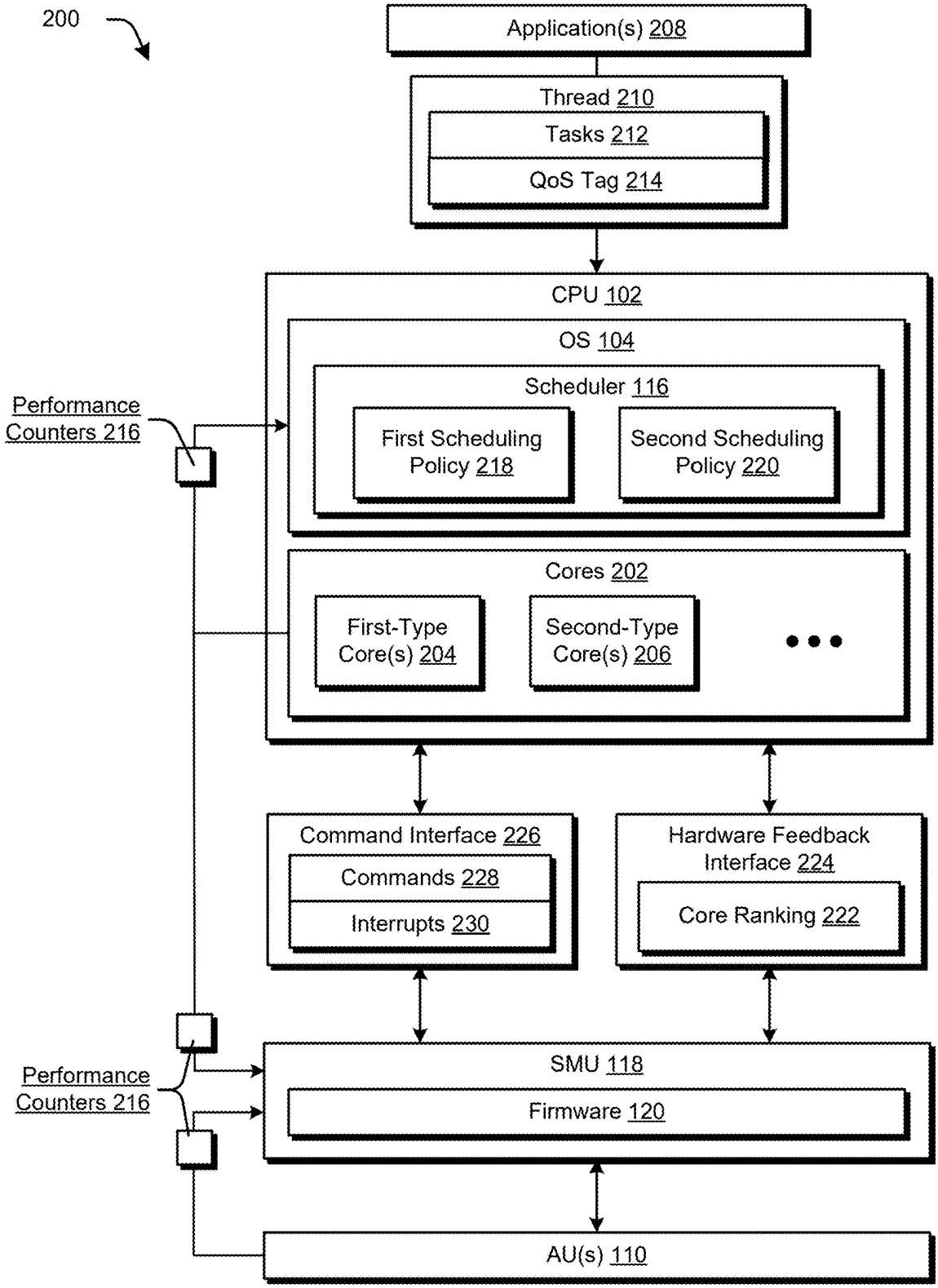

200
Application(s) 208
Thread 210
Tasks 212
QoS Tag 214
CPU 102
OS 104
Scheduler 116
First Scheduling Policy 218
Second Scheduling Policy 220
Performance Counters 216
Cores 202
First-Type Core(s) 204
Second-Type Core(s) 206
Command Interface 226
Commands 228
Interrupts 230
Hardware Feedback Interface 224
Core Ranking 222
Performance Counters 216
SMU 118
Firmware 120
AU(s) 110

FIG. 2

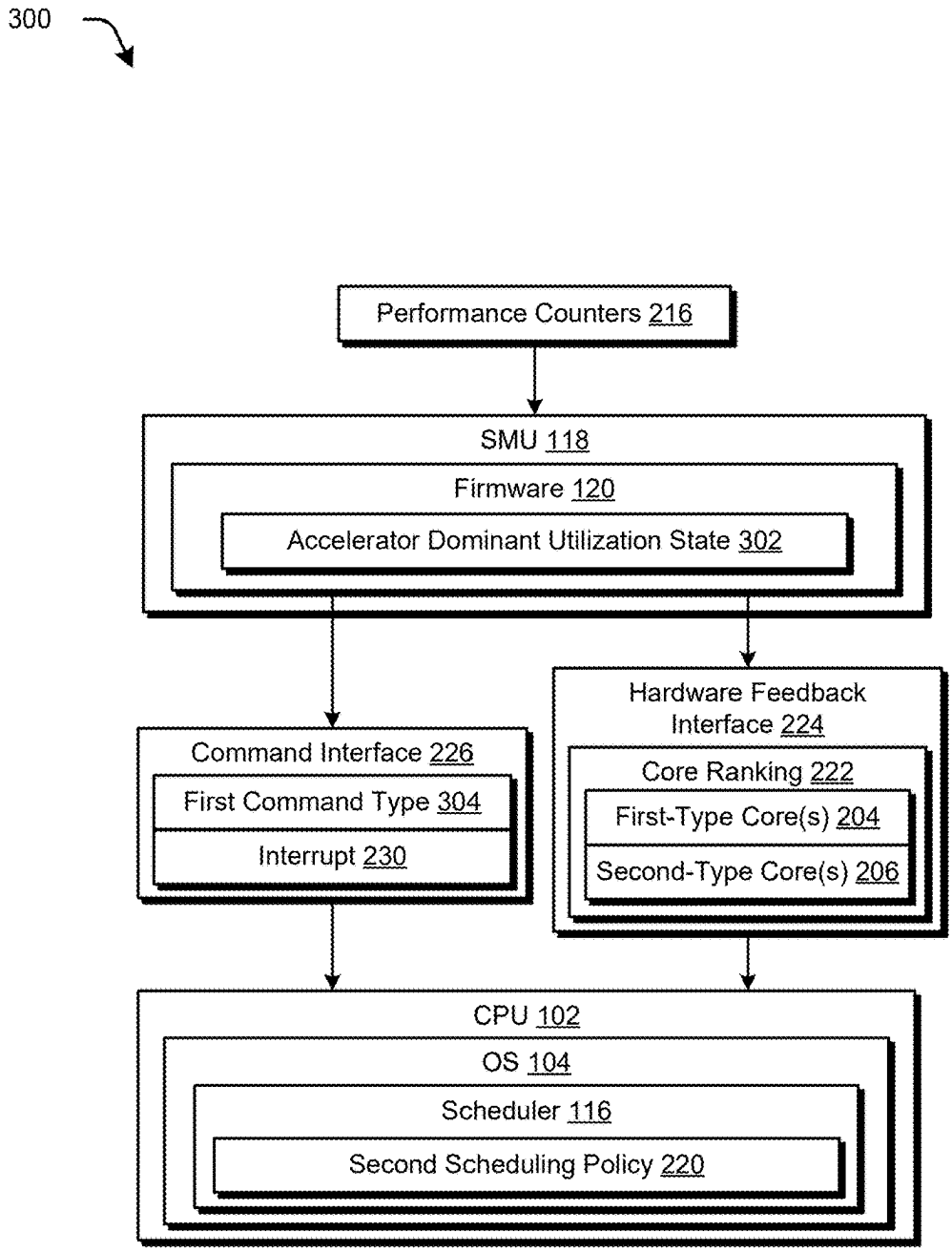

300
Performance Counters 216
SMU 118
Firmware 120
Accelerator Dominant Utilization State 302
Command Interface 226
First Command Type 304
Interrupt 230
Hardware Feedback Interface 224
Core Ranking 222
First-Type Core(s) 204
Second-Type Core(s) 206
CPU 102
OS 104
Scheduler 116
Second Scheduling Policy 220

FIG. 3

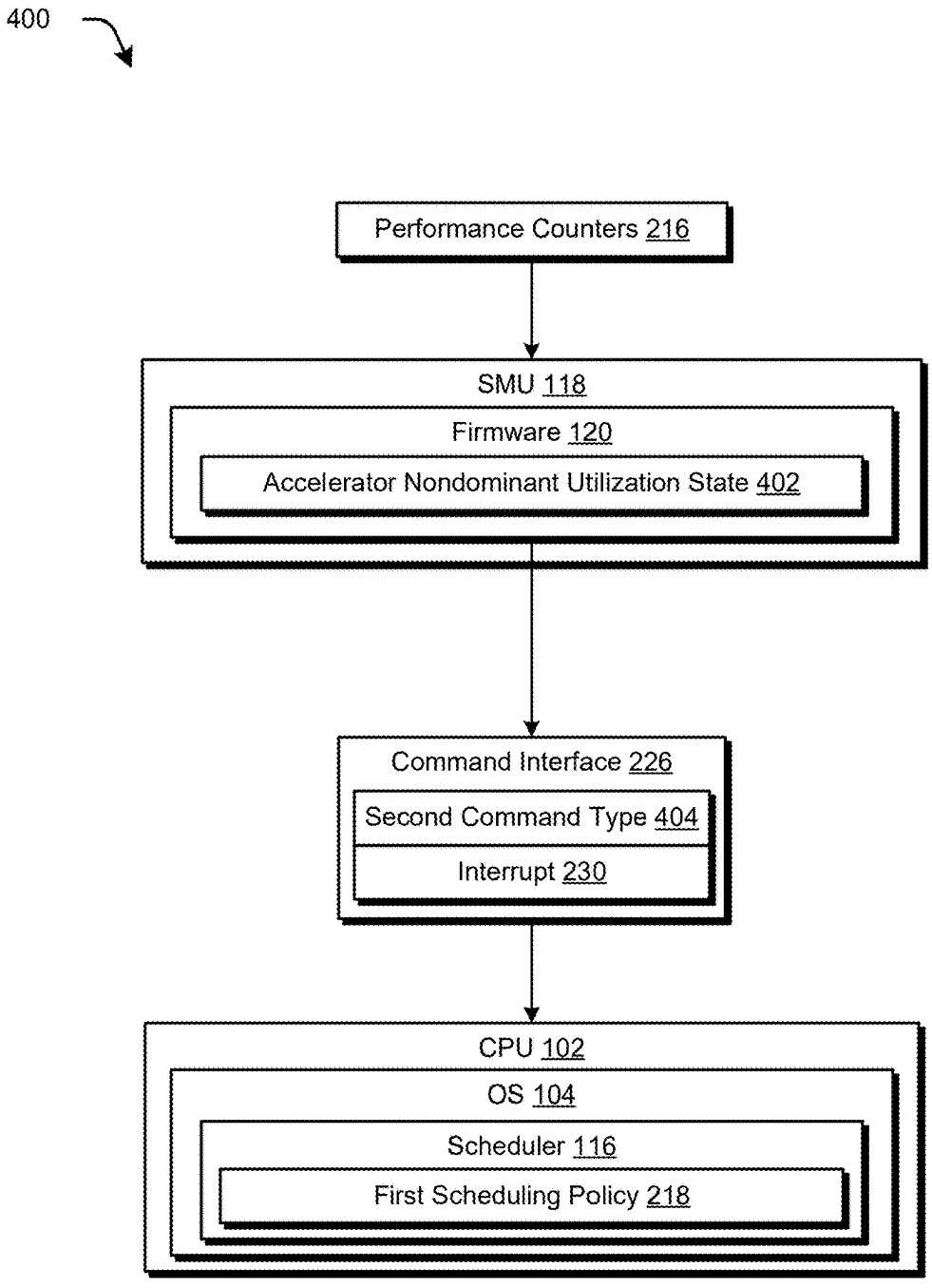

400
Performance Counters 216
SMU 118
Firmware 120
Accelerator Nondominant Utilization State 402
Command Interface 226
Second Command Type 404
Interrupt 230
CPU 102
OS 104
Scheduler 116
First Scheduling Policy 218

FIG. 4

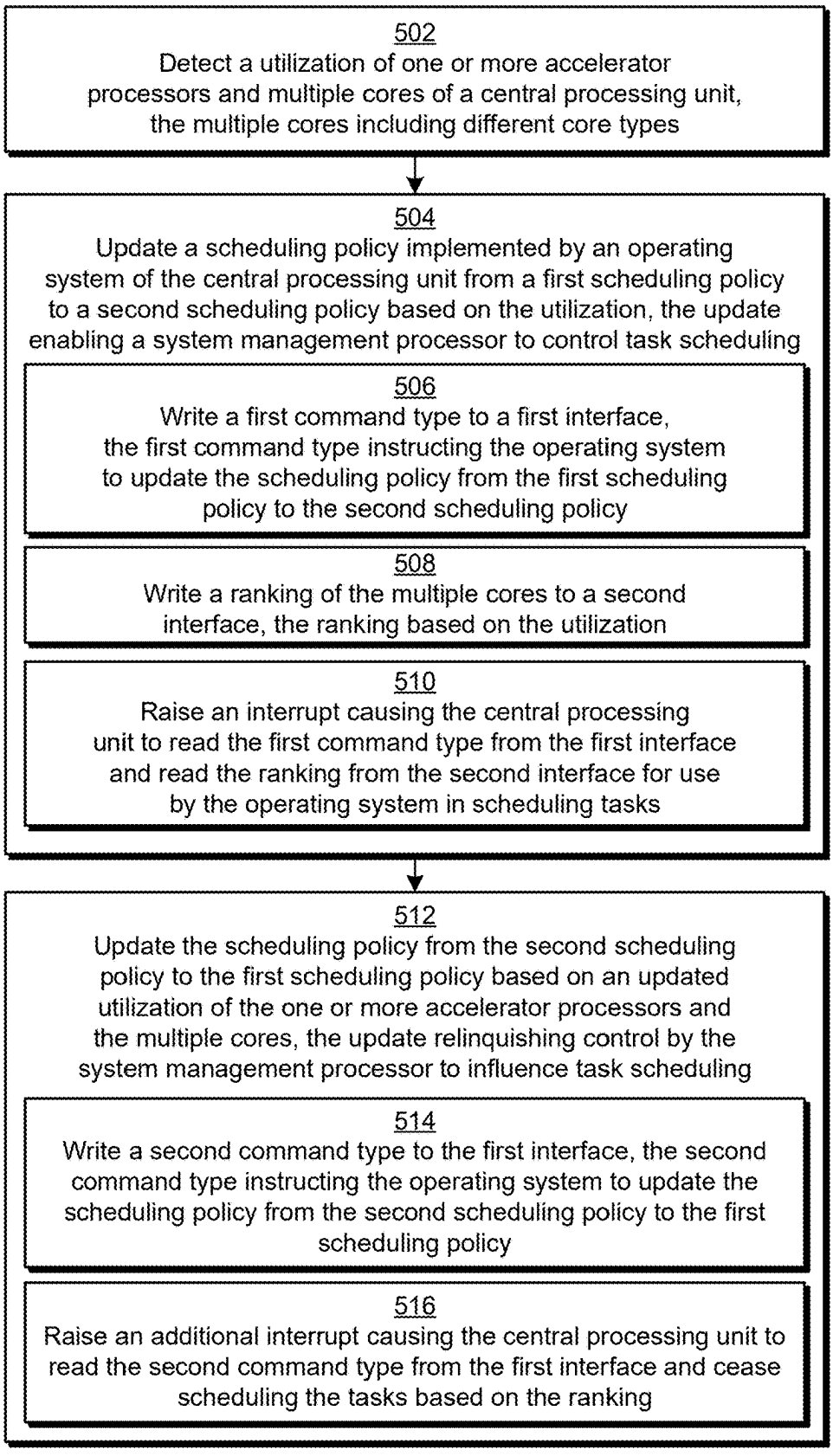

502
Detect a utilization of one or more accelerator processors and multiple cores of a central processing unit, the multiple cores including different core types
504
Update a scheduling policy implemented by an operating system of the central processing unit from a first scheduling policy to a second scheduling policy based on the utilization, the update enabling a system management processor to control task scheduling
506
Write a first command type to a first interface, the first command type instructing the operating system to update the scheduling policy from the first scheduling policy to the second scheduling policy
508
Write a ranking of the multiple cores to a second interface, the ranking based on the utilization
510
Raise an interrupt causing the central processing unit to read the first command type from the first interface and read the ranking from the second interface for use by the operating system in scheduling tasks
512
Update the scheduling policy from the second scheduling policy to the first scheduling policy based on an updated utilization of the one or more accelerator processors and the multiple cores, the update relinquishing control by the system management processor to influence task scheduling
514
Write a second command type to the first interface, the second command type instructing the operating system to update the scheduling policy from the second scheduling policy to the first scheduling policy
516
Raise an additional interrupt causing the central processing unit to read the second command type from the first interface and cease scheduling the tasks based on the ranking

FIG. 5

500

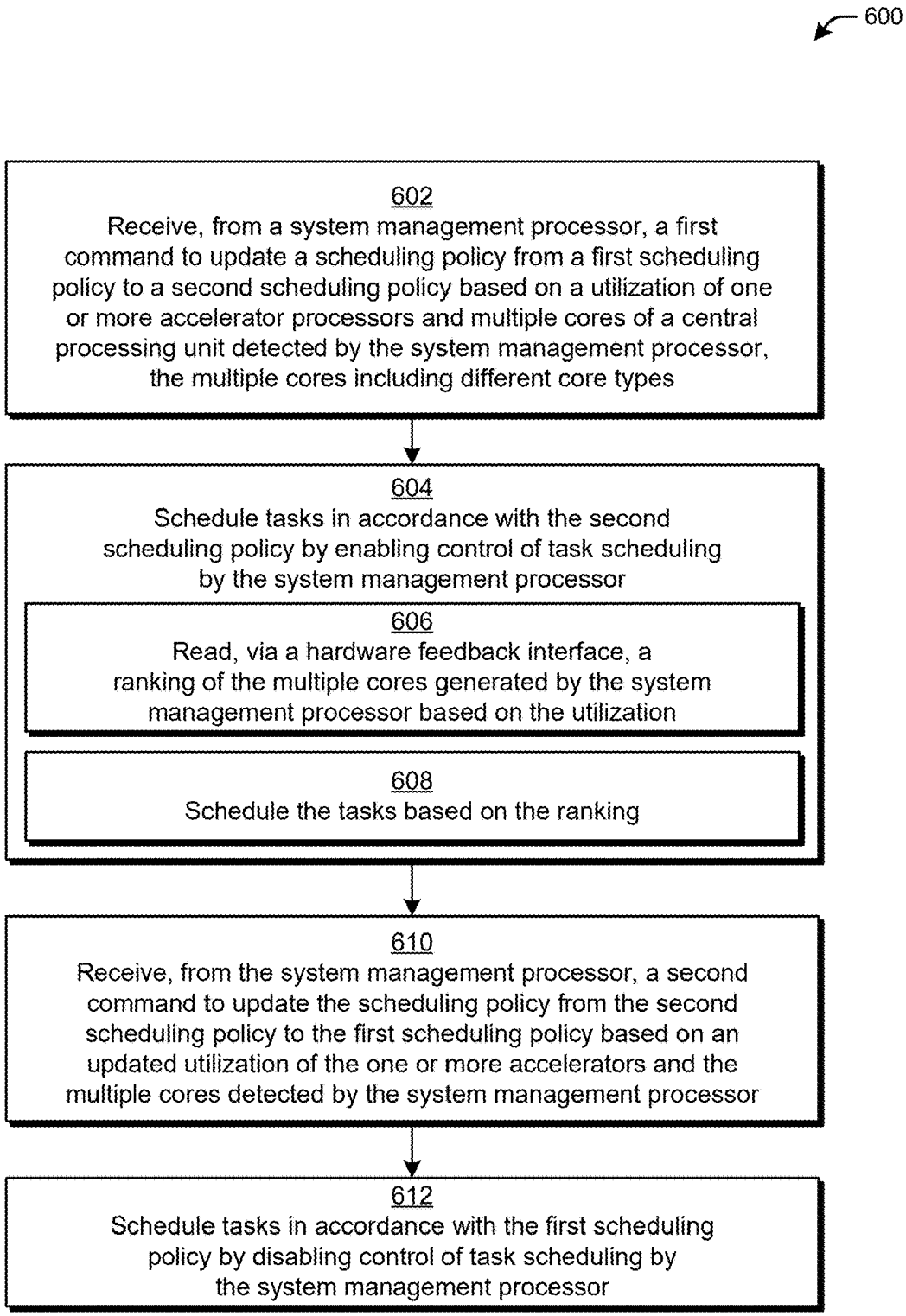

600
602
Receive, from a system management processor, a first command to update a scheduling policy from a first scheduling policy to a second scheduling policy based on a utilization of one or more accelerator processors and multiple cores of a central processing unit detected by the system management processor, the multiple cores including different core types
604
Schedule tasks in accordance with the second scheduling policy by enabling control of task scheduling by the system management processor
606
Read, via a hardware feedback interface, a ranking of the multiple cores generated by the system management processor based on the utilization
608
Schedule the tasks based on the ranking
610
Receive, from the system management processor, a second command to update the scheduling policy from the second scheduling policy to the first scheduling policy based on an updated utilization of the one or more accelerators and the multiple cores detected by the system management processor
612
Schedule tasks in accordance with the first scheduling policy by disabling control of task scheduling by the system management processor

FIG. 6

HYBRID SCHEDULING FOR HETEROGENEOUS PROCESSOR SYSTEMS

RELATED APPLICATION

[0001] This application claims priority under 35 U.S.C. § 119 (e) to U.S. Provisional Patent Application No. 63/561,132, filed Mar. 4, 2024, the entire content of which is hereby incorporated by reference.

BACKGROUND

[0002] Computing systems equipped with multiple types of processors, referred to as heterogeneous processor systems, integrate different types of processors within a single system to optimize performance, power efficiency, and functionality. In a heterogeneous processor system, individual processors (e.g., CPUs) are often equipped with different types of cores, and these processors are referred to as heterogeneous core processors. This architectural design leverages the strengths of various processors and processor cores, such as different core types of a CPU that are either optimized for performance or energy efficiency, graphics processing units (GPUs), and specialized accelerators like digital signal processors (DSPs) or neural processing units (NPUs), to handle different computational tasks effectively. The benefits of such configurations include the ability to dynamically allocate tasks to the most suitable processor and/or core type, significantly enhancing computational efficiency and speed. Another benefit is the ability to utilize energy efficient cores for workloads, tasks, and/or threads that do not require maximal performance, e.g., background tasks.

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] FIG. **1** is a block diagram of a processing system configured to execute one or more applications, in accordance with one or more implementations.

[0004] FIG. **2** is a block diagram of a non-limiting example system to implement techniques of hybrid scheduling for heterogeneous processor systems.

[0005] FIG. **3** depicts a block diagram of a non-limiting example system to transition from a first scheduling policy to a second scheduling policy.

[0006] FIG. **4** depicts a block diagram of a non-limiting example system to transition from a second scheduling policy to a first scheduling policy.

[0007] FIG. **5** depicts a procedure in an example implementation of hybrid scheduling for heterogeneous processing systems as implemented by a system management unit.

[0008] FIG. **6** depicts a procedure in an example implementation of hybrid scheduling for heterogeneous processing systems as implemented by an operating system running on a CPU.

DETAILED DESCRIPTION

[0009] In accordance with the described techniques, a system includes a central processing unit (CPU) having multiple cores of different core types, e.g., the CPU is a heterogeneous core processor. For example, the multiple cores include one or more first-type cores and one or more second-type cores. The first-type cores exhibit increased power efficiencies when ran at reduced clock frequencies (e.g., zero to three GHz), while the second-type cores exhibit increased power efficiencies when ran at increased clock frequencies, e.g., above three GHz. That is, the first-type cores are considered energy efficient cores since running threads on the first-type cores at the reduced clock frequency reduces power consumption for the system. In contrast, the second-type cores are considered performant cores since running threads on the second-type cores at the increased clock frequency increases the speed at which the threads are executed. The system also includes one or more accelerator processors, e.g., GPUs, NPUs, video processing engines (VPEs) and the like. Broadly, the CPU is configured to run an operating system that includes a scheduler, which schedules and dispatches threads for execution by different processing resources of the system, e.g., the cores of the CPU and the accelerator processors.

[0010] In accordance with a first scheduling policy, the scheduler is configured to schedule threads based on quality of service (QOS) tags associated with the threads, which designate priorities (e.g., high priority, medium priority, low priority) associated with the threads. In addition, the scheduler is aware of a degree to which the cores of the CPU are being utilized, but is unaware of a degree to which the accelerator processors are being utilized. Generally, the scheduler prioritizes scheduling threads marked with QoS tags indicating a higher priority for execution by the second-type (e.g., performant) cores, and prioritizes scheduling threads marked with QoS tags indicating a lower priority for execution by the first-type (e.g., energy efficient) cores.

[0011] In accordance with a second scheduling policy, the scheduler is configured to schedule threads based on feedback provided by a system management unit (SMU) via a hardware feedback interface. Here, the SMU periodically reads performance counters from various hardware resources of the system, such as the cores of the CPU and the accelerator processors, which enables the firmware to calculate degrees of utilization of the cores of the CPU and/or the accelerator processors. Based on the degrees of utilization, the firmware communicates feedback to the operating system via the hardware feedback interface, which the scheduler utilizes in making scheduling decisions.

[0012] A goal of thread scheduling in a heterogeneous core CPU, like the one described, is to run lower priority threads on the first-type (e.g., energy efficient) cores to reduce power consumption for the system, while running higher priority threads on the second-type (e.g., performant) cores to increase system responsiveness while executing these threads. The first scheduling policy outperforms the second scheduling policy in achieving this goal in a large majority of utilization scenarios due to the awareness of thread-level processing demands specified by the QoS tags. However, in certain specific scenarios, the second scheduling policy outperforms the first scheduling policy in achieving this goal. These specific scenarios include when the utilization of the CPU cores is low and the utilization of the accelerator processors is high. An example of this scenario is a machine learning workload that is offloaded for execution by NPUs. In these scenarios, the workload that is running on the accelerator processors is the primary, high priority workload, while the threads running on the CPU cores are secondary, lower priority processes. Due to the lack of awareness of the accelerator-level utilization and the inaccuracy of QoS tagging schemes, the first scheduling policy often marks threads as high or medium priority threads during these utilization scenarios. This leads to the scheduler inaccurately dispatching the threads for execution

by the second-type (e.g., performant) cores, which unnecessarily increases power consumption for the system.

[0013] Unlike conventional techniques, the described techniques of hybrid scheduling for heterogeneous processor systems enable a hybrid approach for thread scheduling that dynamically switches between the first scheduling policy and the second scheduling policy. By way of example, the SMU initiates an update from the first scheduling policy to the second scheduling policy responsive to detecting an accelerator dominant utilization state in which the utilization of one or more accelerator processors exceeds an accelerator utilization threshold and the utilization of the CPU cores is below a core utilization threshold. Notably, the update to the second scheduling policy causes the scheduler to use the feedback provided by the SMU via the hardware feedback interface to schedule threads, and cease using the QoS tags to make scheduling decisions. Moreover, the SMU initiates an update from the second scheduling policy to the first scheduling policy responsive to detecting an accelerator nondominant utilization state in which the utilization of the one or more accelerator processors falls below the accelerator utilization threshold and/or the utilization of the CPU cores exceeds the core utilization threshold. The update causes the scheduler to begin scheduling threads based on the QoS tags, and cease using the feedback provided via the hardware feedback interface in making scheduling decisions.

[0014] Here, the described techniques implement a hybrid scheduling approach to transition between the scheduling policies in order to implement the optimal scheduling policy for the utilization scenario. By doing so, the described techniques reduce power consumption for the system while improving or maintaining system responsiveness in executing higher priority threads, as compared to conventional techniques that implement just one scheduling policy.

[0015] In some aspects, the techniques described herein relate to a system including a central processing unit having multiple cores of at least two different core types, one or more accelerator processors, and a system management processor configured to update a scheduling policy implemented by an operating system of the central processing unit from a first scheduling policy to a second scheduling policy based on utilization of the multiple cores and/or the one or more accelerator processors, wherein the update to the second scheduling policy enables the system management processor to control task scheduling.

[0016] In some aspects, the techniques described herein relate to a system, wherein the at least two different core types exhibit different power efficiencies at different clock frequencies.

[0017] In some aspects, the techniques described herein relate to a system, wherein the first scheduling policy includes scheduling tasks based on Quality of Service tags associated with the tasks.

[0018] In some aspects, the techniques described herein relate to a system, wherein to update the scheduling policy, the system management processor is configured to generate a ranking of the multiple cores based on the utilization of the multiple cores and/or the one or more accelerator processors, and communicate the ranking to the operating system for use in scheduling tasks in accordance with the second scheduling policy.

[0019] In some aspects, the techniques described herein relate to a system, wherein the utilization of the multiple cores is below a first threshold and the utilization of the one or more accelerator processors is above a second threshold, and the ranking includes one or more cores of a first core type ranked higher than one or more cores of a second core type, wherein the first core type exhibits increased power efficiency relative to the second core type at a reduced clock frequency.

[0020] In some aspects, the techniques described herein relate to a system, wherein to communicate the ranking, the system management processor is configured to write the ranking to a first interface, write a first command type to a second interface, the first command type instructing the operating system to switch from implementing the first scheduling policy to implementing the second scheduling policy, and raise an interrupt via the second interface causing the central processing unit to read the first command type from the second interface and read the ranking from the first interface for use by the operating system in scheduling the tasks.

[0021] In some aspects, the techniques described herein relate to a system, wherein the system management processor is further configured to write an updated ranking to the first interface based on an updated utilization of the multiple cores, write a third command type to the second interface, the third command type instructing the operating system to continue implementing the second scheduling policy in accordance with the updated ranking, and raise an additional interrupt via the second interface causing the central processing unit to read the third command type from the second interface and read the updated ranking from the first interface for use by the operating system in scheduling the tasks.

[0022] In some aspects, the techniques described herein relate to a system, wherein the system management processor is configured to further update the scheduling policy from the second scheduling policy to the first scheduling policy based on an updated utilization of the multiple cores and/or the one or more accelerator processors, wherein the further update to the first scheduling policy relinquishes control by the system management processor to influence the task scheduling.

[0023] In some aspects, the techniques described herein relate to a system, wherein the updated utilization of the multiple cores is above a first threshold and the updated utilization of the one or more accelerator processors is below a second threshold.

[0024] In some aspects, the techniques described herein relate to a system, wherein to further update the scheduling policy, the system management processor is configured to write a second command type to an interface, the second command type instructing the operating system to switch from implementing the second scheduling policy to implementing the first scheduling policy, and raise an interrupt via the interface causing the central processing unit to read the second command type from the interface.

[0025] In some aspects, the techniques described herein relate to a system management processor configured to detect a utilization of one or more accelerator processors and/or multiple cores of a central processing unit, the multiple cores including at least two different core types, and update a scheduling policy of an operating system from a second scheduling policy to a first scheduling policy based on the utilization, wherein the update to the first scheduling policy relinquishes control by the system management processor to influence task scheduling.

[0026] In some aspects, the techniques described herein relate to a system management processor, wherein the at least two different core types exhibit different power efficiencies at different clock frequencies.

[0027] In some aspects, the techniques described herein relate to a system management processor, wherein the first scheduling policy includes scheduling tasks based on Quality of Service tags associated with the tasks.

[0028] In some aspects, the techniques described herein relate to a system management processor, the second scheduling policy includes scheduling tasks based on a ranking of the multiple cores generated by the system management processor and communicated to the operating system via an interface.

[0029] In some aspects, the techniques described herein relate to a system management processor, wherein the utilization of the multiple cores is above a first threshold and the utilization of the one or more accelerator processors is below a second threshold.

[0030] In some aspects, the techniques described herein relate to a system management processor, wherein to update the scheduling policy, the system management processor is configured to write a second command type to an interface, the second command type instructing the operating system to switch from implementing the second scheduling policy to implementing the first scheduling policy, and raise an interrupt via the interface causing the central processing unit to read the second command type from the interface.

[0031] In some aspects, the techniques described herein relate to a device including a system management processor, one or more accelerator processors, and a central processing unit having multiple cores of at least two different core types, the central processing unit configured to receive, from the system management processor, a command to update a scheduling policy based on utilization of the multiple cores and/or the one or more accelerator processors detected by the system management processor, and update the scheduling policy by enabling or disabling control of task scheduling by the system management processor based on the utilization.

[0032] In some aspects, the techniques described herein relate to a device, wherein the at least two different core types exhibit different power efficiencies at different clock frequencies.

[0033] In some aspects, the techniques described herein relate to a device, wherein the utilization of the multiple cores is below a first threshold and the utilization of the one or more accelerator processors is above a second threshold, and to update the scheduling policy, the central processing unit is configured to enable control of the task scheduling by the system management processor.

[0034] In some aspects, the techniques described herein relate to a device, wherein the utilization of the multiple cores is above a first threshold and the utilization of the one or more accelerator processors is below a second threshold, and to update the scheduling policy, the central processing unit is configured to disable control of the task scheduling by the system management processor.

[0035] FIG. **1** includes a processing system **100** configured to execute one or more applications, such as compute applications (e.g., machine-learning applications, neural network applications, high-performance computing applications, databasing applications, gaming applications), graphics applications, and the like. Examples of devices in which the processing system is implemented include, but are not limited to, a server computer, a personal computer (e.g., a desktop or tower computer), a smartphone or other wireless phone, a tablet or phablet computer, a notebook computer, a laptop computer, a wearable device (e.g., a smartwatch, an augmented reality headset or device, a virtual reality headset or device), an entertainment device (e.g., a gaming console, a portable gaming device, a streaming media player, a digital video recorder, a music or other audio playback device, a television, a set-top box), an Internet of Things (IoT) device, an automotive computer or computer for another type of vehicle, a networking device, a medical device or system, and other computing devices or systems.

[0036] In the illustrated example, the processing system **100** includes a central processing unit (CPU) **102**. In one or more implementations, the CPU **102** is an electronic circuit configured to run an operating system (OS) **104** that manages the execution of applications. For example, the OS **104** is configured to schedule the execution of tasks (e.g., instructions) for applications, allocate portions of resources (e.g., system memory **106**, CPU **102**, input/output (I/O) device **108**, one or more accelerator units (AUs) **110**, storage **114**) for the execution of tasks for the applications, provide an interface to I/O devices (e.g., I/O device **108**) for the applications, or any combination thereof. In particular, the OS **104** includes a scheduler **116** (e.g., implemented in software of the OS **104** kernel) that schedules the execution of tasks (e.g., instructions) for applications, such as by dispatching the tasks among different processing resources of the processing system **100** (e.g., different cores of the CPU **102** and/or different AUs **110**) for execution.

[0037] In this example, the processing system **100** includes a system management unit (SMU) **118** which is communicatively coupled to the CPU **102** (e.g., via the connection circuitry **130**) and the AU **110** via one or more wired or wireless connections. In general, the SMU **118** is an electronic circuit (e.g., a processor or a controller) configured to perform operations related to power, thermal, and hardware management for the processing system **100**. To carry out such operations, the SMU **118** includes firmware **120** (e.g., power management firmware), which is executable code or software instructions embedded within (e.g., stored in non-volatile memory of) the SMU **118**. Broadly, the SMU **118** (by running the firmware **120**) is configured to control power states, adjust voltages and power delivery, monitor temperature sensors, activate cooling mechanisms (e.g., fans), and manage clock frequencies for circuitry components of the processing system **100** such as the CPU **102** (and cores thereof), the AU **110**, and the memory **106**. In accordance with the described techniques, the SMU **118** (by running the firmware **120**) is configured to update a task scheduling policy implemented by the scheduler **116** based on a detected utilization of the processing system **100**, e.g., utilization of the CPU **102** and the AUs **110**.

[0038] Here, the SMU **118** and the firmware **120** are implemented as separate components of the processing system **100**. In variations, however, the SMU **118** and/or the firmware **120** are included in and/or implemented by one or more different components of the processing system **100**, such as the CPU **102**, the memory **106**, the I/O device **108**, the AU **110**, the I/O circuitry **112**, the storage **114**, and so forth. In at least one implementation, the SMU **118** and the firmware **120** or portions of the SMU **118** and the firmware **120** are included in at least two of the depicted components

of the processing system **100**. By way of example, the SMU **118** and the firmware **120** may be included in or otherwise implemented by at least the CPU **102**, the AU **110**, and the I/O circuitry **112**.

[0039] The CPU **102** includes one or more processor chiplets **122**, which are communicatively coupled together by a data fabric **124** in one or more implementations. Each of the processor chiplets **122**, for example, includes one or more processor cores **126**, **128** configured to concurrently execute one or more series of instructions, also referred to herein as "threads," for an application. Further, the data fabric **124** communicatively couples each processor chiplet **122**-N of the CPU **102** such that each processor core (e.g., processor cores **126**) of a first processor chiplet (e.g., **116-1**) is communicatively coupled to each processor core (e.g., processor cores **128**) of one or more other processor chiplets **122**. Though the example embodiment presented in FIG. **1** shows a first processor chiplet (**122-1**) having three processor cores (**126-1**, **126-2**, **126**-K) representing a K number of processor cores **126** and a second processor chiplet (**122**-N) having three processor cores (e.g., **128-1**, **128-2**, **128**-L) representing an L number of processor cores **128**, in other implementations (L being an integer number greater than or equal to one), each processor chiplet **122** may have any number of processor cores **126**, **128**. For example, each processor chiplet **122** can have the same number of processor cores **126**, **128** as one or more other processor chiplets **122**, a different number of processor cores **126**, **128** as one or more other processor chiplets **122**, or both. Examples of connections which are usable to implement data fabric include but are not limited to, buses (e.g., a data bus, a system, an address bus), interconnects, memory channels, through silicon vias, traces, and planes. Other example connections include optical connections, fiber optic connections, and/or connections or links based on quantum entanglement.

[0040] Additionally, within the processing system **100**, the CPU **102** is communicatively coupled to an I/O circuitry **112** by a connection circuitry **130**. For example, each processor chiplet **122** of the CPU **102** is communicatively coupled to the I/O circuitry **112** by the connection circuitry **130**. In addition, the SMU **118** is communicatively coupled to the CPU **102** by the connection circuitry **130**. The connection circuitry **130** includes, for example, one or more data fabrics, buses, buffers, queues, and the like. The I/O circuitry **112** is configured to facilitate communications between two or more components of the processing system **100** such as between the CPU **102**, system memory **106**, display **132**, universal serial bus (USB) devices, peripheral component interconnect (PCI) devices (e.g., I/O device **108**, AU **110**), storage **114**, the SMU **118**, and the like.

[0041] As an example, system memory **106** includes any combination of one or more volatile memories and/or one or more non-volatile memories, examples of which include dynamic random-access memory (DRAM), static random-access memory (SRAM), non-volatile RAM, and the like. To manage access to the system memory **106** by CPU **102**, the I/O device **108**, the AU **110**, and/or any other components, the I/O circuitry **112** includes one or more memory controllers **134**. These memory controllers **134**, for example, include circuitry configured to manage and fulfill memory access requests issued from the CPU **102**, the I/O device **108**, the AU **110**, or any combination thereof. Examples of such requests include read requests, write requests, fetch requests, pre-fetch requests, or any combination thereof. That is to say, these memory controllers **134** are configured to manage access to the data stored at one or more memory addresses within the system memory **106**, such as by CPU **102**, the I/O device **108**, and/or the AU **110**.

[0042] When an application is to be executed by processing system **100**, the OS **104** running on the CPU **102** is configured to load at least a portion of program code **136** (e.g., an executable file) associated with the application from, for example, a storage **114** into system memory **106**. This storage **114**, for example, includes a non-volatile storage such as a flash memory, solid-state memory, hard disk, optical disc, or the like configured to store program code **136** for one or more applications.

[0043] To facilitate communication between the storage **114** and other components of processing system **100**, the I/O circuitry **112** includes one or more storage connectors **138** (e.g., universal serial bus (USB) connectors, serial AT attachment (SATA) connectors, PCI Express (PCIe) connectors) configured to communicatively couple storage **114** to the I/O circuitry **112** such that I/O circuitry **112** is capable of routing signals to and from the storage **114** to one or more other components of the processing system **100**.

[0044] In association with executing an application, in one or more scenarios, the CPU **102** is configured to issue one or more instructions (e.g., threads) to be executed for an application to the AU **110**. The AU **110** is configured to execute these instructions by operating as one or more vector processors, coprocessors, graphics processing units (GPUs), general-purpose GPUs (GPGPUs), video encoding/decoding processors, video processing engines (VPEs), display rendering processors, non-scalar processors, highly parallel processors, artificial intelligence (AI) processors (also known as neural processing units, or NPUs), inference engines, machine-learning processors, other multithreaded processing units, scalar processors, serial processors, programmable logic devices (e.g., field-programmable logic devices (FPGAs)), or any combination thereof.

[0045] In at least one example, the AU **110** includes one or more compute units that concurrently execute one or more threads of an application and store data resulting from the execution of these threads in AU memory **140**. This AU memory **140**, for example, includes any combination of one or more volatile memories and/or non-volatile memories, examples of which include caches, video RAM (VRAM), or the like. In one or more implementations, these compute units are also configured to execute these threads based on the data stored in one or more physical registers **142** of the AU **110**.

[0046] To facilitate communication between the AU **110** and one or more other components of processing system **100**, the I/O circuitry **112** includes or is otherwise connected to one or more connectors, such as PCI connectors **144** (e.g., PCIe connectors) each including circuitry configured to communicatively couple the AU **110** to the I/O circuitry such that the I/O circuitry **112** is capable of routing signals to and from the AU **110** to one or more other components of the processing system **100**. Further, the PCIe connectors **144** are configured to communicatively couple the I/O device **108** to the I/O circuitry **112** such that the I/O circuitry **112** is capable of routing signals to and from the I/O device **108** to one or more other components of the processing system **100**.

[0047] By way of example and not limitation, the I/O device **108** includes one or more keyboards, pointing devices, game controllers (e.g., gamepads, joysticks), audio input devices (e.g., microphones), touch pads, printers, speakers, headphones, optical mark readers, hard disk drives, flash drives, solid-state drives, and the like. Additionally, the I/O device **108** is configured to execute one or more operations, tasks, instructions, or any combination thereof based on one or more physical registers **146** of the I/O device **108**. In one or more implementations, such physical registers **146** are configured to maintain data (e.g., operands, instructions, values, variables) indicating one or more operations, tasks, or instructions to be performed by the I/O device **108**.

[0048] To manage communication between components of the processing system **100** (e.g., AU **110**, I/O device **108**) that are connected to PCI connectors **144**, and one or more other components of the processing system **100**, the I/O circuitry **112** includes PCI switch **148**. The PCI switch **148**, for example, includes circuitry configured to route packets to and from the components of the processing system **100** connected to the PCI connectors **144** as well as to the other components of the processing system **100**. As an example, based on address data indicated in a packet received from a first component (e.g., CPU **102**), the PCI switch **148** routes the packet to a corresponding component (e.g., AU **110**) connected to the PCI connectors **144**.

[0049] Based on the processing system **100** executing a graphics application, for instance, the CPU **102**, the AU **110**, or both are configured to execute one or more instructions (e.g., draw calls) such that a scene including one or more graphics objects is rendered. After rendering such a scene, the processing system **100** stores the scene in the storage **114**, displays the scene on the display **132**, or both. The display **132**, for example, includes a cathode-ray tube (CRT) display, liquid crystal display (LCD), light emitting diode (LED) display, organic light emitting diode (OLED) display, or any combination thereof. To enable the processing system **100** to display a scene on the display **132**, the I/O circuitry **112** includes display circuitry **150**. The display circuitry **150**, for example, includes high-definition multimedia interface (HDMI) connectors, DisplayPort connectors, digital visual interface (DVI) connectors, USB connectors, and the like, each including circuitry configured to communicatively couple the display **132** to the I/O circuitry **112**. Additionally or alternatively, the display circuitry **150** includes circuitry configured to manage the display of one or more scenes on the display **132** such as display controllers, buffers, memory, or any combination thereof.

[0050] Further, the CPU **102**, the AU **110**, or both are configured to concurrently run one or more virtual machines (VMs), which are each configured to execute one or more corresponding applications. To manage communications between such VMs and the underlying resources of the processing system **100**, such as any one or more components of processing system **100**, including the CPU **102**, the I/O device **108**, the AU **110**, and the system memory **106**, the I/O circuitry **112** includes memory management unit (MMU) **152** and input-output memory management unit (IOMMU) **154**. The MMU **152** includes, for example, circuitry configured to manage memory requests, such as from the CPU **102** to the system memory **106**. For example, the MMU **152** is configured to handle memory requests issued from the CPU **102** and associated with a VM running on the CPU **102**. These memory requests, for example, request access to read, write, fetch, or pre-fetch data residing at one or more virtual addresses (e.g., guest virtual addresses) each indicating one or more portions (e.g., physical memory addresses) of the system memory **106**. Based on receiving a memory request from the CPU **102**, the MMU **152** is configured to translate the virtual address indicated in the memory request to a physical address in the system memory **106** and to fulfill the request.

[0051] The IOMMU **154** includes, for example, circuitry configured to manage memory requests (memory-mapped I/O (MMIO) requests) from the CPU **102** to the I/O device **108**, the AU **110**, or both, and to manage memory requests (direct memory access (DMA) requests) from the I/O device **108** or the AU **110** to the system memory **106**. For example, to access the registers **146** of the I/O device **108**, the registers **142** of the AU **110**, and/or the AU memory **140**, the CPU **102** issues one or more MMIO requests. MMIO requests each request access to read, write, fetch, or pre-fetch data residing at one or more virtual addresses (e.g., guest virtual addresses) which each represent at least a portion of the registers **146** of the I/O device **108**, the registers **142** of the AU **110**, or the AU memory **140**, respectively. As another example, to access the system memory **106** without using the CPU **102**, the I/O device **108**, the AU **110**, or both are configured to issue one or more DMA requests. Such DMA requests each request access to read, write, fetch, or pre-fetch data residing at one or more virtual addresses (e.g., device virtual addresses) which each represent at least a portion of the system memory **106**. Based on receiving an MMIO request or DMA request, the IOMMU **154** is configured to translate the virtual address indicated in the MMIO or DMA request to a physical address and fulfill the request.

[0052] In variations, the processing system **100** can include any combination of the components depicted and described. For example, in at least one variation, the processing system **100** does not include one or more of the components depicted and described in relation to FIG. **1**. Additionally or alternatively, in at least one variation, the processing system **100** includes additional and/or different components from those depicted. The processing system **100** is configurable in a variety of ways with different combinations of components in accordance with the described techniques.

[0053] FIG. **2** is a block diagram of a non-limiting example system **200** to implement techniques of hybrid scheduling for heterogeneous processor systems. The system **200** includes the CPU **102** running the OS **104** and the scheduler **116**, the SMU **118** running the firmware **120**, and one or more AUs **110**, as depicted and described above with reference to FIG. **1**. In accordance with the described techniques, the CPU **102**, the SMU **118**, and the one or more AUs **110** are coupled to one another via one or more wired and/or wireless connections. Example wired connections include, but are not limited to, buses (e.g., a data bus), interconnects, traces, and planes.

[0054] Here, the CPU **102** includes a plurality of cores **202** of different types, e.g., the CPU **102** is a heterogenous core processor. By way of example, the cores **202** include one or more first-type cores **204** and one or more second-type cores **206**. Generally, the different core types implemented by the CPU **102** exhibit increased power efficiencies at different clock frequencies. Notably, power efficiency of a processor

or processor core is a ratio of performance (e.g., measured in instructions executed per second) to power consumption, e.g., measured in watts. Power efficiency can be expressed mathematically as

$$\text{Power Efficiency} = \frac{\text{Performance}}{\text{Power Consumption}}.$$

Clock frequency, as applied to a processor or processor core, is the number of processing cycles that the processor or processor core executes per second, e.g., measured in hertz (Hz).

[0055] Here, the first-type core **204** exhibits increased power efficiency relative to the second-type core **206** at a reduced clock frequency. For example, the first-type core **204** exhibits increased power efficiency relative to the second-type core **206** at clock frequencies between zero and three GHz, while the second-type core **206** exhibits increased power efficiency relative to the first-type core **204** at clock frequencies that are above three GHz. That is, the first-type cores **204** are more energy efficient than the second-type cores **206** since executing threads or tasks on the first-type cores **204** at the reduced clock frequency reduces power consumption for the system. In contrast, the second-type cores **206** are considered more performant than the first-type cores **204** since executing tasks or threads on the second-type cores **206** at the increased clock frequency increases the speed at which the tasks or threads are executed. In various examples performance of a core **202** is measured in instructions executed per processor cycle (e.g., IPC), and the maximum IPC achievable by the second-type core **206** is greater than the maximum IPC achievable by the first-type core **204**. Although two core types of the CPU **102** are depicted and described herein, it is to be appreciated that the cores **202** include three or more types of cores each exhibiting increased power efficiencies at different ranges of clock frequencies.

[0056] The AUs **110** are specialized processors implemented in electronic circuitry to execute certain tasks and/or workloads with increased computational efficiency relative to the CPU **102**. By way of example, the AUs **110** include a GPU configured to execute graphics processing tasks faster than the CPU **102**, an NPU and/or an inference processor configured to execute machine learning and/or artificial intelligence workloads faster than the CPU **102**, a VPE and/or video encoding/decoding processor configured to execute video processing (e.g., encoding and decoding) workloads faster than the CPU **102**, and/or display rendering processors configured to render display data and/or manage display outputs faster than the CPU **102**.

[0057] In other words, the system **200** is representative of a heterogeneous processor system including a plurality of different processor types (e.g., the CPU **102** and the AUs **110**) and different core types, e.g., the first-type cores **204** and the second-type cores **206** of the CPU **102**. In accordance with this heterogeneous processor configuration, the scheduler **116** is configured to schedule tasks for execution by the most suitable processor and/or core type. In one example, this includes dispatching specialized workloads for execution by corresponding AUs **110** (e.g., dispatching machine learning workloads for execution by NPUs and/or inference processors) to speed up execution of the specialized workloads. Additionally or alternatively, the scheduler **116** dispatches high priority tasks for execution by the second-type cores **206** (e.g., which can execute more instructions per cycle than the first-type cores **204**) to speed up execution of the high priority tasks. In yet another non-limiting example, the scheduler **116** dispatches low priority tasks for execution by the first-type cores **204** to reduce power consumption for the system **200**.

[0058] As previously mentioned, the OS **104** is configured to manage execution of one or more applications **208**. As part of this, the applications **208** submit threads **210** of tasks **212** to be executed by the CPU **102** and/or the AUs **110**. Here, tasks **212** are software instructions executable by one or more processors or processor cores. Further, threads **210** are series or sequences of tasks **212** (e.g., software instructions) that are executable by one or more processors or processor cores. In various examples, the applications **208** implements multithreading, which is the ability of an application **208** to submit multiple threads that are executable concurrently on different processors (e.g., the CPU **102** and/or the AUs **110**) or different processor cores (e.g., the cores **202**). Multithreading enables parallel computing of multiple threads, which speeds up execution of the application **208**.

[0059] As shown, the threads **210** are marked with Quality of Service (QOS) tags **214**. By way of example, an application **208** marks a thread **210** with a QoS tag **214** indicative of a priority for scheduling the thread **210**. In a specific but non-limiting example, threads **210** are markable with "high priority," "medium priority," and "low priority" QoS tags **214**. In this example, the high priority threads **210** are allocated more hardware (e.g., processing, memory, and communication) resources, and/or are scheduled for more immediate execution as compared to medium and low priority threads **210**. Further, low priority threads **210** are allocated less hardware (e.g., processing, memory, and communication) resources, and/or are delayed to give preference to high and medium priority threads **210**. Moreover, medium priority threads are allocated more hardware resources than low priority threads but less hardware resources than high priority threads. Additionally or alternatively, medium priority threads **210** are delayed to give preference to high priority threads **210**, but are scheduled for more immediate execution as compared to low priority threads **210**.

[0060] Hardware resources expected to be invoked (e.g., processing, memory, communication, and I/O resources) by a thread **210** impact the thread's QoS tag **214**. For example, a degree to which a thread **210** is compute-bound, memory-bound, I/O bound, and communication-bound impacts the thread's QoS tag **214**. Additionally or alternatively, the context in which the thread **210** is running impacts the thread's QoS tag **214**. By way of example, the threads **210** running the following processes are associated with progressively higher priority QoS tags: (1) background processes (e.g., data indexing and OS **104** updates), (2) user-initiated processes triggered by user input (e.g., opening a file), (3) user-interactive processes (e.g., displaying a user interface with which the user is actively engaging).

[0061] As part of managing execution of the applications **208**, the OS **104** includes the scheduler **116** that schedules the threads **210** for execution by different processing resources, e.g., the cores **202** and/or the AUs **110**. As part of this, the scheduler **116** receives a thread **210** and determines which core **202** or which AU **110** to execute the thread **210**. Further, the scheduler **116** dispatches the thread **210** to the

determined core **202** or the determined AU **110** for execution. As discussed herein, operations performed by the OS **104** or the scheduler **116** are operations performed by the CPU **102** by running the OS **104** or the scheduler **116**.

[0062] As previously mentioned, the SMU **118** (by running the firmware **120**) is configured to perform operations related to power and thermal management of system circuitry components. As part of its power and thermal management functionality, the firmware **120** is configured to receive performance counters **216** from each of the cores **202** and each of the AUs **110**. For example, each core **202** and each AU **110** maintains performance counters **216** (e.g., within hardware registers) that track performance metrics associated with respective cores **202** and respective AUs **110**. Periodically (e.g., every millisecond), the cores **202** and the AUs **110** update the performance counters **216**, and the firmware **120** reads the updated performance counters **216**. Examples of performance metrics tracked by the performance counters **216** include a total number of instructions executed during a certain time period and IPC. Notably, operations described herein as performed by the firmware **120**, are operations performed by the SMU **118** by running the firmware **120**.

[0063] Based on the performance counters **216**, the firmware **120** calculates the utilization of the cores **202** and the AUs **110**. Taking IPC as an example performance metric conveyable by the performance counters **216**, the firmware **120** includes indications of maximum IPCs achievable by individual processing resources, e.g., individual cores **202** and AUs **110**. Given this, the firmware **120** receives a performance counter **216** indicative of an actual IPC currently implemented by a processing resource (e.g., a core **202** or AU **110**), and the firmware **120** calculates the utilization of the processing resource using the actual IPC and the maximum IPC based on the following relationship:

$$\text{Utilization} = \frac{\text{Actual } IPC}{\text{Maximum } IPC}.$$

This process is repeated for each processing resource to determine the utilization of each individual core **202** and each individual AU **110**. More generally, the utilization of a processing resource is the percentage of total processing capability of the processing resource that is currently being utilized.

[0064] As shown, the scheduler **116** includes a first scheduling policy **218** and a second scheduling policy **220**. Broadly, a scheduling policy includes or corresponds to a set of rules, instructions, and/or criteria defining when and how threads **210** are scheduled for execution. In accordance with the first scheduling policy **218**, the scheduler **116** schedules threads **210** based on the QoS tags **214**, utilization of the cores **202**, and one or more operating conditions of the system **200**. By way of example, the OS **104** also receives the performance counters **216** from the cores **202** (but not the AUs **110**), and the OS **104** calculates the utilization of the cores **202** in a similar manner to the firmware **120**. Examples of the operating conditions include a user-tunable parameter that specifies a balance between competing scheduling preferences for the OS **104**, such as application responsiveness vs. throughput, power efficiency vs. performance, and the like, e.g., also referred to as a "slider position." Additional examples of the operating conditions include remaining battery life and whether the device is connected to an external power source (e.g., receiving alternating current (AC) power from an outlet) or an internal power source, e.g., receiving direct current (DC) power from an internal battery.

[0065] In accordance with the first scheduling policy **218**, the scheduler **116** generally schedules threads **210** with higher priority QoS tags **214** for execution by the second-type cores **206**. By doing so, the second-type cores **206** can be run at increased clock frequencies (e.g., at which the second-type cores **206** are more power efficient) to take advantage of the increased IPC achievable by the second-type core **206**, which increases the speed at which the high priority threads **210** are executed. In addition, the scheduler **116** generally schedules threads **210** with lower priority QoS tags **214** for execution by the first-type cores **202**. By doing so, the first-type cores **204** can be run at reduced clock frequencies (e.g., at which the first-type cores **204** are more power efficient), thereby reducing power consumption for the system **200**.

[0066] The operating conditions and utilization of the cores **202** also impact task scheduling in accordance with the first scheduling policy **218**. In various examples, the scheduler **116** prioritizes dispatching threads to cores **202** having lower levels of utilization. Additionally or alternatively, the scheduler **116** prioritizes dispatching threads **210** to first-type cores **204** when the remaining battery life is low, the device is not connected to an external AC power source, the user-tunable parameter specifies a preference of power efficiency over performance, and so on. Contrarily, the scheduler **116** prioritizes dispatching threads **210** to second-type cores **206** when battery life is high and/or the device is connected to an external AC power source, the user-tunable parameter specifies a preference of performance over power efficiency, and so on.

[0067] In accordance with the second scheduling policy **220**, the scheduler **116** is configured to schedule the threads **210** based on a core ranking **222** of the cores **202** generated by the firmware **120**. Unlike the OS **104**, the firmware **120** receives the performance counters **216** of the AUs **110** in addition to the performance counters **216** of the cores **202**, and therefore, the utilization of the cores **202** and the AUs **110** is available to the firmware **120**. Based on the utilization of the cores **202** and the AUs, the firmware **120** detects a utilization scenario of the system **200** in which to override the first scheduling policy **218** in order for the firmware **120** to control task/thread scheduling.

[0068] In various examples, this utilization scenario is a scenario in which the utilization of the cores **202** is below a core utilization threshold, and the utilization of one or more of the AUs **110** is above an AU utilization threshold. This utilization scenario is indicative of a specialized workload being executed by one or more AUs **110** designed to execute the specialized workloads, e.g., the system **200** is executing a machine learning workload on one or more NPUs and/or inference processors. During the execution of these workloads, it is preferable for the threads **210** that are executing on the cores **202** of the CPU **102** be executed on the first-type cores **204** at the reduced clock frequency in order to reduce power consumption for the system **200**. This is because, while the primary, specialized workload is running on one or more AUs **110**, the threads **210** that are running on the CPU **102** are secondary and/or lower priority processes. However, QoS tagging schemes often inaccurately mark threads **210** with high or medium priority QoS tags **214**

during execution of these workloads, leading to the threads **210** being executed on the second-type cores **206**. Given the inaccuracy of QoS tagging schemes as well as the inability of the OS **104** to detect the utilization of the AUs **110**, the firmware **120** is configured to initiate an update to cause the scheduler **116** to switch to the second scheduling policy **220**.

[0069] To do so, the firmware **120** generates the core ranking **222** based on the detected utilization scenario. When the utilization scenario is indicative of a high degree of AU **110** utilization (e.g., that exceeds the AU utilization threshold) and a low degree of CPU core **202** utilization (e.g., that falls below the core utilization threshold), for instance, the first-type cores **204** are ranked higher than the second-type cores **206** in the core ranking **222**. Furthermore, the firmware **120** writes the core ranking **222** to a hardware feedback interface **224**, which is a system that provides feedback to the OS **104** and/or scheduler **116** for use in making thread/task scheduling decisions. In one or more implementations, the hardware feedback interface **224** includes registers or memory locations accessible by the OS **104**. Given this, the firmware **120** writes the core ranking **222** to the registers or memory locations (e.g., via data buses or communication channels) and the OS **104** reads the core ranking **222** from the registers or memory locations, e.g., via data buses or communication channels. In one or more examples, the registers or memory locations include a data structure (e.g., a core ranking table) to which the core ranking **222** is written.

[0070] In accordance with the second scheduling policy **220**, the scheduler **116** schedules threads **210** and/or tasks **212** based on the core ranking **222**, e.g., by prioritizing dispatching the threads **210** and/or tasks **212** to first-type cores **204** which are ranked higher in the core ranking **222**. In one or more implementations, the scheduler **116** does not rely on the QoS tags **214** when making scheduling decisions in using the second scheduling policy **220**. In other words, the firmware **120** controls task/thread scheduling via the core ranking **222** provided by the hardware feedback interface **224** in accordance with the second scheduling policy **220**.

[0071] In various implementations, the firmware **120** is configured to initiate updates and/or changes to the scheduling policy of the scheduler **116** between the first scheduling policy **218** and the second scheduling policy **220**. To do so, the system includes a command interface **226**, which is an interface enabling the OS **104** to communicate directly with the firmware **120**. In one specific but non-limiting example, the command interface **226** is a platform communications channel (PCC), as described in the Advanced Configuration and Power Interface (ACPI) Specification which is hereby incorporated by reference. In particular, the ACPI specification describes a Type 4 Slave Subspace, which is used by the platform to send asynchronous notifications to the OS **104** for Operating System Power Management (OSPM) tasks. Here, the command interface **226** is implemented as a Type 4 Slave Subspace of a PCC, and the command interface **226** is used by the firmware **120** to send commands **228** and interrupts **230** to the OS **104** for the purpose of directing the OS **104** and scheduler **116** to transition between the first scheduling policy **218** and the second scheduling policy **220**.

[0072] In response to detecting the utilization scenario in which to override the first scheduling policy **218** and enable the firmware **120** to control thread/task scheduling, the firmware **120** writes a command **228** of a first command type to the command interface **226**. The first command type instructs the OS **104** and scheduler **116** to transition from the first scheduling policy **218** to the second scheduling policy **220**. After having written the core ranking **222** to the hardware feedback interface **224**, the firmware **120** additionally raises an interrupt **230** via the command interface **226**. Broadly, the interrupt **230** is a flag or other indication conveying that a command **228** has been written to the command interface **226** and is ready to be executed by the OS **104**. Thus, responsive to the interrupt **230** being raised, the OS **104** reads the first command type from the command interface **226**, reads the core ranking **222** from the hardware feedback interface **224**, and the scheduler **116** transitions to the second scheduling policy **220** to schedule threads **210** in accordance with the core ranking **222**.

[0073] In response to detecting a usage scenario in which to transition back to the first scheduling policy **218** and relinquish control by the firmware **120** to influence thread/task scheduling, the firmware **120** writes a command **228** of a second command type to the command interface **226**. The second command type instructs the OS **104** and scheduler **116** to transition from the second scheduling policy **220** to the first scheduling policy **218**. In addition, the firmware raises an interrupt **230** via the command interface **226**. Responsive to the interrupt **230** being raised, the OS **104** reads the second command type and the scheduler **116** transitions to the first scheduling policy **218**, in part, by no longer relying on the core ranking **222** for making scheduling decisions. Regardless of the scheduling policy **218**, **220**, the scheduler **116** dispatches threads **210** to the AUs **110** based on the threads **210** having been designated (e.g., marked) by the applications **208** for execution by the AUs **110**.

[0074] Here, unlike conventional scheduling techniques, the described techniques relate to a hybrid scheduling paradigm in which the OS **104** and the scheduler **116** transition between a first scheduling policy **218** and a second scheduling policy **220** based on utilization of the cores **202** and the AUs **110**. While the first scheduling policy **218** has the advantage of scheduling threads **210** based on software thread-level processing demands using the QoS tags **214**, the first scheduling policy **218** does not account for the utilization of the AUs **110**. Moreover, while the second scheduling policy **220** has the advantage of scheduling threads **210** based on the utilization of the platform as a whole (e.g., including the cores **202** and the AUs **110**), the second scheduling policy **220** does not account for the software thread-level processing demands. Due to this, the first and second scheduling policies **218**, **220** are more effective in different usage scenarios.

[0075] Indeed, one goal of task scheduling in a heterogeneous processor system is to reduce power consumption by running lower priority threads **210** on the more energy efficient first-type cores **204** and increasing performance and system responsiveness for higher priority threads **210** by running such threads **210** on the more performant second-type cores **206**. In most scenarios, the first scheduling policy **218** outperforms the second scheduling policy **220** in terms of achieving this goal due to the scheduling decisions being made on the basis of the thread-level processing demands specified by the QoS tags **214**. However, in certain specific scenarios such as the aforementioned utilization scenario in which the utilization of the AUs **110** exceeds the utilization

of the cores **202**, the second scheduling policy **220** outperforms the first scheduling policy in terms of achieving the goal. This is because, due to the reliance on the QoS tags **214** and the unawareness of the AU **110** utilization, the first scheduling policy **218** inaccurately schedules threads **210** for execution on the second-type cores **206** despite the threads **210** running processes that are secondary to the primary process that is running on the AUs **110**, Here, the described techniques implement a hybrid scheduling approach to transition between the scheduling policies **218**, **220** in order to implement the optimal scheduling policy for the utilization scenario. By doing so, the described techniques reduce power consumption for the system **200** while improving or maintaining performance of higher priority threads **210**, as compared to conventional single policy scheduling techniques.

[0076] FIG. **3** depicts a block diagram of a non-limiting example system **300** to transition from a first scheduling policy to a second scheduling policy. Here, the firmware **120** receives the performance counters **216** from the cores **202** and the AUs **110**. In particular, the firmware **120** reads the performance counters **216** from hardware registers of the cores **202** and the AUs **110**. Based on the performance counters **216**, the firmware **120** calculates a utilization of each individual core **202** and/or each individual AU **110** in accordance with the described techniques. Furthermore, the firmware **120** detects an accelerator dominant utilization state **302** based on the calculated utilizations.

[0077] More specifically, the firmware **120** detects an accelerator dominant utilization state **302** based on the utilization of the cores **202** falling below a core utilization threshold and/or the utilization of one or more AUs **110** exceeding an AU utilization threshold, e.g., which can be the same or different for different types of AUs **110**. The utilization of the cores **202** is a function of the number of cores **202** that are being utilized as well as the degree of utilization calculated for the individual cores. In one or more implementations, the utilization of one or more NPUs and/or inference processors exceeds a threshold utilization, and as such, the accelerator dominant utilization state **302** is indicative of a machine learning workload running on the processor system. Additionally or alternatively, the utilization of one or more VPEs, video encoding/decoding processors, and/or display rendering processors exceeds a utilization threshold, e.g., the accelerator dominant utilization state is indicative of a video conferencing workload running on the processor system. Notably, the thresholds defining the accelerator dominant utilization state **302** are programmed as part of the firmware **120** in various implementations.

[0078] Notably, utilization of processors and processor cores is also a measure of memory-boundedness and compute-boundedness of the workload that is running on the cores **202**. For example, higher degrees of processor or core utilization are indicative of higher degrees of compute boundedness, while lower degrees of processor or core utilization are indicative of higher degrees of memory boundedness. Thus, the accelerator dominant utilization state **302** is also conceptualizable as a workload that is compute-bound on the AU **110** but not compute-bound (e.g., memory-bound) on the CPU **102**.

[0079] Responsive to detecting the accelerator dominant utilization state **302**, the firmware **120** generates the core ranking **222** and writes the core ranking **222** to the hardware feedback interface **224**. When the accelerator dominant utilization state **302** is detected, the first-type cores **204** are ranked higher than the second-type cores **206** as shown, thereby directing the OS **104** to prioritize dispatching threads **210** for execution by the first-type cores **204** over the second-type cores **206**. Also responsive to detecting the accelerator dominant utilization state **302**, the firmware **120** writes a command **228** of a first command type **304** to the command interface **226**. The command **228** of the first command type **304** is an instruction directing the OS **104** and the scheduler **116** to switch from implementing the first scheduling policy **218** to implementing the second scheduling policy **220**. In other words, the command **228** of the first command type **304** is an instruction to begin using feedback and/or hints provided by the firmware **120** via the core ranking **222** in making scheduling decisions. After the first command type **304** is written to the command interface **226** and the core ranking **222** is written to the hardware feedback interface **224**, the firmware **120** raises an interrupt **230** via the command interface **226**.

[0080] Responsive to the interrupt **230** being raised, the OS **104** reads the first command type **304** from the command interface **226**. Based on the first command type **304**, the OS **104** reads the core ranking **222** from the hardware feedback interface **224**. Although not shown, the OS **104** stores a copy of the core ranking **222** in a memory resource (e.g., a cache system or hardware registers) of the CPU **102** in various implementations. By doing so, the OS **104** and/or the scheduler **116** can access the core ranking **222** from the CPU memory resource, which is faster than accessing the core ranking **222** from the hardware feedback interface **224**.

[0081] Based on the first command type **304**, the scheduler **116** transitions from the first scheduling policy **218** to the second scheduling policy **220**, as shown. As part of this, the firmware **120** controls task/thread scheduling via the core ranking **222**. For instance, the scheduler **116** ceases to use the QoS tags **214**, the operating conditions, and the core **202** utilization when making scheduling decision, and begins using the core ranking table to schedule threads **210**. In accordance with the second scheduling policy **220**, for instance, the scheduler **116** prioritizes scheduling threads **210** for execution by the first-type cores **204** over the second-type cores **206** based on the core ranking **222**. As part of this, the scheduler **116** migrates currently executing threads **210** from the second-type cores **206** to the first-type cores **204**, and the scheduler **116** dispatches newly received threads **210** for execution by the first-type cores **204** (if available) rather than the second-type core **206**. In various scenarios, implementation of the second scheduling policy **220** involves dispatching threads **210** that are marked with high priority QoS tags **214** for execution by the first-type (e.g., energy efficient) cores **206**.

[0082] FIG. **4** depicts a block diagram of a non-limiting example system **400** to transition from a second scheduling policy to a first scheduling policy. Here, the firmware **120** receives the performance counters **216** from the cores **202** and the AUs **110**. In particular, the firmware **120** reads the performance counters **216** from hardware registers of the cores **202** and the AUs **110**. Based on the performance counters **216**, the firmware **120** calculates a utilization of each individual core **202** and/or each individual AU **110** in accordance with the described techniques. Furthermore, the firmware **120** detects an accelerator nondominant utilization state **402** based on the calculated utilizations. In particular, the firmware **120** detects the accelerator nondominant utili-

zation state **402** based on the utilization of the of the cores **202** exceeding the core utilization threshold, and/or the utilization of one or more AUs **110** falling below the AU utilization threshold.

[0083] Responsive to detecting the accelerator nondominant utilization state **402**, the firmware **120** writes a command **228** of a second command type **404** to the command interface **226**. The command **228** of the second command type **404** is an instruction directing the OS **104** and the scheduler **116** to switch from implementing the second scheduling policy **220** to implementing the first scheduling policy **218**. In other words, the command **228** of the second command type **404** is an instruction to cease using feedback and/or hints provided by the firmware **120** via the core ranking **222** in making scheduling decisions. After the second command type **404** is written to the command interface **226**, the firmware **120** raises an interrupt **230** via the command interface **226**.

[0084] Responsive to the interrupt **230** being raised, the OS **104** reads the second command type **404** from the command interface **226**. Based on the second command type **404**, the scheduler **116** transitions from the second scheduling policy **220** to the first scheduling policy **218**, as shown. As part of this, the firmware **120** relinquishes control of task/thread scheduling via the core ranking **222**. For instance, the scheduler **116** ceases to use the core ranking **222** when making scheduling decision, and begins using the QoS tags **214**, the operating conditions, and the core **202** utilization to schedule threads **210**.

[0085] While examples are described herein in which transitions between the first scheduling policy **218** and the second scheduling policy **220** are initiated based on utilization of the cores **202** and the AUs **110**, these examples are not to be construed as limiting. Rather, the firmware **120** overrides the first scheduling policy **218** to implement the second scheduling policy **220** in any scenario detectable by the firmware **120** in which the system would benefit from ignoring the QoS tags **214** and using feedback from the firmware **120** (e.g., from the hardware platform) in making scheduling decisions. Examples of these scenarios include power throttling scenarios and scenarios in which there is a high thermal condition (e.g., overheating) or other error conditions, e.g., memory errors and CPU faults.

[0086] In one or more implementations, the firmware **120** is configured to initiate a transition to the second scheduling policy **220** responsive to detecting these scenarios. By way of example, the firmware **120** generates a core ranking **222** that ranks the first-type cores **204** higher than the second-type cores **206** and communicates the ranking to the OS **104** and the scheduler **116** using the hardware feedback interface **224** and the command interface **226**. As a result, the firmware **120** controls the thread scheduling via the core ranking **222**. Responsive to detecting the termination of these scenarios, the firmware **120** also initiates a transition back to the first scheduling policy **218**. By way of example, the firmware **120** writes the second command type **404** to the command interface **226** and raises the interrupt **230**, which causes the scheduler **116** to schedule threads **210** in accordance with the first scheduling policy **218**. In other words, the firmware **120** relinquishes control of thread scheduling responsive to detecting a termination of the aforementioned scenarios.

[0087] In addition to or as part of the specific scenarios described above, the firmware **120** is configured to initiate a transitions between the first scheduling policy **218** and the second scheduling policy **220** based on any one or any combination of a variety of factors. These factors include, but are not limited to including, utilization of the cores **202** of the CPU **102**, utilization of the AUs **110**, compute-boundedness and memory-boundedness of a workload executing on the CPU **102** and/or the AUs **110**, a number of threads **210** executing on the CPU **102** and/or the AUs **110**, a degree of parallelism exhibited by the threads **210** running on the CPU **102** and/or the AUs **110**, a degree of complexity exhibited by the threads **210** running on the CPU and/or the AUs **110**, thermal conditions of hardware resources of the processing system **100**, whether power throttling conditions are applied to hardware resources of the processing system **100**, and various error conditions, such as memory errors and CPU faults.

[0088] In various examples, the firmware **120** is configured to update the core ranking **222** while the scheduler **116** is implementing the second scheduling policy **220**. In accordance with the described techniques, the firmware **120** determines to update the core ranking **222** based on a detected utilization of the cores **202**. By way of example, the firmware **120** detects that the utilization of a highest-ranked core **202** of the core ranking **222** exceeds a threshold, and as such, the firmware **120** determines to reduce the utilization of the core **202**. To do so, the firmware **120** generates an updated core ranking **222** that rearranges an ordering of the first-type cores **204** and writes the updated core ranking **222** to the hardware feedback interface **224**. Moreover, the firmware **120** writes a command **228** of a third command type to the command interface **226**. The third command type instructs the scheduler **116** to continue implementing the second scheduling policy **220** in accordance with the updated core ranking **222**. Once the updated core ranking **222** has been written to the hardware feedback interface **224** and the third command type has been written to the command interface **226**, the firmware **120** raises an interrupt **230**. The interrupt **230** causes the OS **104** to read the third command type from the command interface **226**, read the updated core ranking **222** from the hardware feedback interface **224**, and continue implementing the second scheduling policy **220** by scheduling threads **210** based on the updated core ranking. In one or more implementations, the OS **104** updates the copy of the core ranking **222** in the CPU **102** memory resource to include the updated core ranking **222**.

[0089] While the example scenarios involve ranking the first-type cores **204** higher than the second-type cores **206** in the core ranking **222**, these examples are not to be construed as limiting. Rather, the firmware **120** ranks the cores **202** in any order depending on the utilization scenario, including ranking the first-type cores **204** higher than the second-type cores **206**, ranking the second-type cores **206** higher than the first-type cores **204**, and interspersing the first-type cores **204** and the second-type cores **206** within the core ranking **222**.

[0090] FIG. **5** depicts a procedure **500** in an example implementation of hybrid scheduling for heterogeneous processing systems as implemented by a system management unit. In the procedure **500**, a utilization of one or more accelerator processors and multiple cores of a central processing unit is detected, and the multiple cores include different core types (block **502**). By way of example, the firmware **120** detects the utilization of one or more AUs and

the cores **202** of the CPU **102**. Notably, the cores **202** include first-type cores **204** and second-type cores **206**, and the first-type cores **202** exhibit increased power efficiency relative to the second-type cores **206** at reduced clock frequencies.

[0091] A scheduling policy implemented by an operating system of the central processing unit is updated from a first scheduling policy to the second scheduling policy based on the utilization, and the update enables a system management processor to control task scheduling (block **504**). For example, the firmware **120** initiates a transition that causes the scheduler **116** to change from scheduling tasks in accordance with the first scheduling policy **218** to scheduling tasks in accordance with the second scheduling policy **220**. The transition is initiated based on the utilization of one or more AUs **110** exceeding an AU utilization threshold, and the utilization of the cores **202** falling below a core utilization threshold, e.g., the accelerator dominant utilization state **302**. Broadly, the second scheduling policy **220** enables the firmware **120** to control and/or influence task scheduling.

[0092] As part of updating the scheduling policy, a first command type is written to a first interface, and the first command type instructs the operating system to update the scheduling policy from the first scheduling policy to the second scheduling policy (block **506**). For instance, the firmware **120** writes a command **228** of the first command type **304** to the command interface **226**, and the first command type **304** is an instruction to change from the first scheduling policy **218** to the second scheduling policy **220**.

[0093] As part of updating the scheduling policy, a ranking of the multiple cores is written to a second interface, and the ranking is based on the utilization (block **508**). Based on the detected utilization, for instance, the firmware **120** generates a core ranking **222** which ranks the first-type cores **204** higher than the second-type cores **206**. Furthermore, the firmware **120** writes the core ranking **222** to the hardware feedback interface **224**.

[0094] As part of updating the scheduling policy, an interrupt is raised that causes the central processing unit to read the first command type from the first interface and read the ranking from the second interface for use by the operating system in scheduling tasks (block **510**). By way of example, the firmware **120** raises an interrupt **230** via the command interface **226** after having written the first command type **304** and the core ranking **222** to the command interface **226** and the hardware feedback interface **224**, respectively. Responsive to the interrupt **230** being raised, the OS **104** reads the first command type **304** from the command interface **226**. Furthermore, the OS **104** reads the core ranking **222** from the hardware feedback interface **224** based on the first command type **304**. Furthermore, the scheduler **116** implements the second scheduling policy **220** by scheduling threads **210** based on the core ranking **222**, while ignoring the QoS tags **214**.

[0095] The scheduling policy is updated from the second scheduling policy to the first scheduling policy based on an updated utilization of the one or more accelerator processors and the multiple cores, and the update relinquishes control by the system management processor to influence task scheduling (block **512**). For example, the firmware **120** detects an updated utilization of the one or more AUs **110** and the cores **202**. Here, the utilization of the one or more AUs **110** is below the AU utilization threshold and/or the utilization of the cores **202** is above the core utilization threshold, e.g., the accelerator nondominant utilization state **402**. Based on the detected utilization, the firmware **120** initiates a transition that causes the scheduler **116** to change from scheduling tasks in accordance with the second scheduling policy **220** to scheduling tasks in accordance with the first scheduling policy **218**. Broadly, task scheduling in accordance with the first scheduling policy **218** is uncontrolled and/or uninfluenced by the firmware **120**.

[0096] As part of updating the scheduling policy, a second command type is written to the first interface, and the second command type instructs the operating system to update the scheduling policy from the second scheduling policy to the first scheduling policy (block **514**). Here, the firmware **120** writes a command **228** of the second command type **404** to the command interface **226**, and the second command type **404** is an instruction to change from the second scheduling policy **220** to the first scheduling policy **218**.

[0097] As part of updating the scheduling policy, an additional interrupt is raised that causes the central processing unit to read the second command type from the first interface and cease scheduling the tasks based on the ranking (block **516**). For instance, the firmware **120** raises an interrupt **230** via the command interface **226** after having written the second command type **404** to the command interface **226**. Responsive to the interrupt **230** being raised, the OS **104** reads the second command type **404** from the command interface **226**. Based on the second command type **404**, the scheduler **116** implements the first scheduling policy **218** by scheduling threads **210** based on the QoS tags **214**, and no longer scheduling threads **210** based on the core ranking **222**.

[0098] FIG. **6** depicts a procedure **600** in an example implementation of hybrid scheduling for heterogeneous processing systems as implemented by an operating system running on a CPU. In the procedure **600**, a first command is received from a system management processor to update a scheduling policy from a first scheduling policy to a second scheduling policy based on a utilization of one or more accelerator processors and multiple cores of a central processing unit detected by the system management processor, and the multiple cores include different core types (block **602**). By way of example, the firmware **120** detects the utilization of one or more AUs **110** and the cores **202** (e.g., including the first-type cores **204** and the second-type cores **206**) of the CPU **102**. In particular, the firmware **120** detects the accelerator dominant utilization state **302**. In response, the firmware **120** writes a command **228** of the first command type **304** to the command interface **226** and raises an interrupt **230** via the command interface **226**. The OS **104**, in response to the interrupt **230** being raised, reads the first command type **304** from the command interface **226**. Notably, the first command type **304** is an instruction to transition from scheduling tasks in accordance with the first scheduling policy **218** to scheduling tasks in accordance with the second scheduling policy **220**.

[0099] Tasks are scheduled in accordance with the second scheduling policy by enabling control of task scheduling by the system management processor (block **604**). Based on the first command type **304**, the scheduler **116** begins scheduling tasks in accordance with the second scheduling policy **220**. Broadly, the second scheduling policy **220** enables the firmware **120** to control and/or influence task scheduling.

[0100] As part of implementing the second scheduling policy, a ranking of the cores generated by the system

management processor based on the utilization is read via a hardware feedback interface (block **606**). For instance, the OS **104** reads the core ranking **222** from the hardware feedback interface **224** based on the first command type **304**. Notably, the core ranking **222** is generated by the firmware **120** based on the utilization of the cores **202** and the one or more AUs **110**.

[0101] As part of implementing the second scheduling policy, tasks are scheduled based on the ranking (block **608**). For example, the scheduler **116** implements the second scheduling policy **220** by scheduling threads **210** based on the core ranking **222**, while ignoring the QoS tags **214**.

[0102] A second command is received from the system management processor to update the scheduling policy from the second scheduling policy to the first scheduling policy based on an updated utilization of the one or more accelerators and the multiple cores detected by the system management processor (block **610**). Here, the firmware **120** detects an updated utilization of one or more AUs **110** and the cores **202**. In particular, the firmware **120** detects the accelerator nondominant utilization state **402**. In response, the firmware **120** writes a command **228** of the second command type **404** to the command interface **226** and raises an interrupt **230** via the command interface **226**. The OS **104**, in response to the interrupt **230** being raised, reads the second command type **404** from the command interface **226**. Notably, the second command type **404** is an instruction to transition from scheduling tasks in accordance with the second scheduling policy **220** to scheduling tasks in accordance with the first scheduling policy **218**.

[0103] Tasks are scheduled in accordance with the first scheduling policy by disabling control of task scheduling by the system management processor (block **612**). Based on the second command type **404**, the scheduler **116** begins scheduling tasks in accordance with the first scheduling policy **218**. Broadly, task scheduling in accordance with the first scheduling policy **218** is uncontrolled and/or uninfluenced by the firmware **120**. More specifically, the scheduler **116** implements the first scheduling policy **218** by scheduling threads **210** based on the QoS tags **214**, and no longer scheduling threads **210** based on the core ranking **222**.

What is claimed is:

1. A system comprising:

a central processing unit having multiple cores of at least two different core types;

one or more accelerator processors; and

a system management processor configured to update a scheduling policy implemented by an operating system of the central processing unit from a first scheduling policy to a second scheduling policy based on utilization of the multiple cores and/or the one or more accelerator processors, wherein the update to the second scheduling policy enables the system management processor to control task scheduling.

2. The system of claim **1**, wherein the at least two different core types exhibit different power efficiencies at different clock frequencies.

3. The system of claim **1**, wherein the first scheduling policy includes scheduling tasks based on Quality of Service tags associated with the tasks.

4. The system of claim **1**, wherein to update the scheduling policy, the system management processor is configured to:

generate a ranking of the multiple cores based on the utilization of the multiple cores and/or the one or more accelerator processors; and

communicate the ranking to the operating system for use in scheduling tasks in accordance with the second scheduling policy.

5. The system of claim **4**, wherein the utilization of the multiple cores is below a first threshold and the utilization of the one or more accelerator processors is above a second threshold, and the ranking includes one or more cores of a first core type ranked higher than one or more cores of a second core type, wherein the first core type exhibits increased power efficiency relative to the second core type at a reduced clock frequency.

6. The system of claim **4**, wherein to communicate the ranking, the system management processor is configured to:

write the ranking to a first interface;

write a first command type to a second interface, the first command type instructing the operating system to switch from implementing the first scheduling policy to implementing the second scheduling policy; and

raise an interrupt via the second interface causing the central processing unit to read the first command type from the second interface and read the ranking from the first interface for use by the operating system in scheduling the tasks.

7. The system of claim **6**, wherein the system management processor is further configured to:

write an updated ranking to the first interface based on an updated utilization of the multiple cores;

write a third command type to the second interface, the third command type instructing the operating system to continue implementing the second scheduling policy in accordance with the updated ranking; and

raise an additional interrupt via the second interface causing the central processing unit to read the third command type from the second interface and read the updated ranking from the first interface for use by the operating system in scheduling the tasks.

8. The system of claim **1**, wherein the system management processor is configured to further update the scheduling policy from the second scheduling policy to the first scheduling policy based on an updated utilization of the multiple cores and/or the one or more accelerator processors, wherein the further update to the first scheduling policy relinquishes control by the system management processor to influence the task scheduling.

9. The system of claim **8**, wherein the updated utilization of the multiple cores is above a first threshold and the updated utilization of the one or more accelerator processors is below a second threshold.

10. The system of claim **8**, wherein to further update the scheduling policy, the system management processor is configured to:

write a second command type to an interface, the second command type instructing the operating system to switch from implementing the second scheduling policy to implementing the first scheduling policy; and

raise an interrupt via the interface causing the central processing unit to read the second command type from the interface.

11. A system management processor configured to:

detect a utilization of one or more accelerator processors and/or multiple cores of a central processing unit, the multiple cores including at least two different core types; and

update a scheduling policy of an operating system from a second scheduling policy to a first scheduling policy based on the utilization, wherein the update to the first scheduling policy relinquishes control by the system management processor to influence task scheduling.

12. The system management processor of claim **11**, wherein the at least two different core types exhibit different power efficiencies at different clock frequencies.

13. The system management processor of claim **11**, wherein the first scheduling policy includes scheduling tasks based on Quality of Service tags associated with the tasks.

14. The system management processor of claim **11**, the second scheduling policy includes scheduling tasks based on a ranking of the multiple cores generated by the system management processor and communicated to the operating system via an interface.

15. The system management processor of claim **11**, wherein the utilization of the multiple cores is above a first threshold and the utilization of the one or more accelerator processors is below a second threshold.

16. The system management processor of claim **11**, wherein to update the scheduling policy, the system management processor is configured to:

write a second command type to an interface, the second command type instructing the operating system to switch from implementing the second scheduling policy to implementing the first scheduling policy; and

raise an interrupt via the interface causing the central processing unit to read the second command type from the interface.

17. A device comprising:

a system management processor;

one or more accelerator processors; and

a central processing unit having multiple cores of at least two different core types, the central processing unit configured to:

receive, from the system management processor, a command to update a scheduling policy based on utilization of the multiple cores and/or the one or more accelerator processors detected by the system management processor; and

update the scheduling policy by enabling or disabling control of task scheduling by the system management processor based on the utilization.

18. The device of claim **17**, wherein the at least two different core types exhibit different power efficiencies at different clock frequencies.

19. The device of claim **17**, wherein the utilization of the multiple cores is below a first threshold and the utilization of the one or more accelerator processors is above a second threshold, and to update the scheduling policy, the central processing unit is configured to enable control of the task scheduling by the system management processor.

20. The device of claim **17**, wherein the utilization of the multiple cores is above a first threshold and the utilization of the one or more accelerator processors is below a second threshold, and to update the scheduling policy, the central processing unit is configured to disable control of the task scheduling by the system management processor.

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